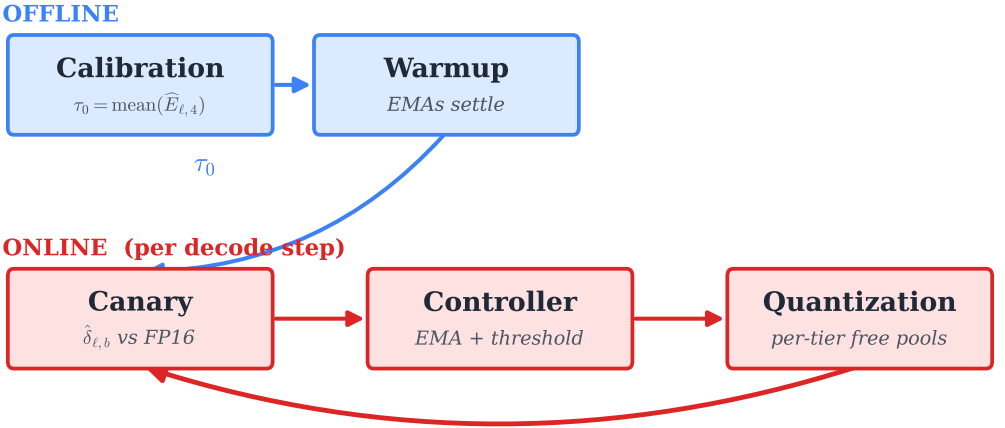
OFFLINE
Calibration
$\tau_0 = \mathrm{mean}(\widehat{E}_{\ell,4})$
Warmup
EMAs settle
τ_0
ONLINE (per decode step)
Canary
$\hat{\delta}_{\ell,b}$ vs FP16
Controller
EMA + threshold
Quantization
per-tier free pools
feedback: new K/V blocks observed by canary